CoxyInsure DApp

Front End Design Document

Project CoxyInsure DeFi Insurance DApp

Document Type Front End Design Document

Purpose Describe the user-facing interface, layout strategy, screen flows, reusable UI components, and admin-facing control surfaces.

Prepared For Product, Design, and Development Review

Version 1.0

This document follows the structure of the provided front-end reference template, adapted to the CoxyInsure insurance workflow and current application screens.

Table of Contents

1. Introduction
2. Design Goals and Principles
3. Front-End Information Architecture
4. Global Layout and Navigation
5. Public Entry Experience
  5.1 Landing / Home Screen
  5.2 Wallet Connection and Account Access
6. Dashboard Overview
7. Membership Experience
8. Pools and Liquidity Management View
9. Buy Cover Experience
10. Claims Submission and Claim History
11. Governance Voting and Execution
12. Admin Front-End Surfaces
13. History, Audit, and Search Views
14. Security, Wallet Binding, and Status Feedback
15. Reusable UI Components
16. Responsive Behaviour
17. Accessibility and Usability Notes
18. Future Front-End Enhancements
19. Summary

1. Introduction

The CoxyInsure front end is designed to let members connect a wallet, contribute liquidity, buy protocol cover, submit claims with evidence, participate in community governance, and allow approved administrators to review and execute claim outcomes from a coordinated interface.

The interface follows a professional financial dashboard pattern: a clear header, persistent navigation, action-oriented cards, strong state feedback, and structured tables that make on-chain insurance actions understandable to both technical and non-technical users.

2. Design Goals and Principles

- Make blockchain insurance actions feel familiar by using standard dashboard, card, and form patterns.
- Keep critical actions visible: connect wallet, mint membership, deposit liquidity, buy cover, submit claim, vote, and review history.
- Present trust signals clearly through wallet identity, member verification, claim evidence links, and governance status indicators.
- Reduce user error with locked premium tiers, labeled inputs, validation feedback, and explicit status messaging.
- Maintain a consistent visual system across user pages and the separate admin portal.
- Support desktop-first use while preserving readability on smaller laptop and tablet-width layouts.

3. Front-End Information Architecture

The front end is organized around the following major user-visible areas:

- Public landing and entry orientation
- Wallet connection and session establishment
- Membership onboarding and verification
- Pools and liquidity dashboard
- Buy cover experience
- Claims submission and personal claim history
- Governance voting and claim execution visibility
- Audit/history and transaction review
- Admin dashboard, active covers, claims, governance, and membership views
- Status, modal, and copy-to-clipboard feedback layers

Placeholder A. Landing / public entry screenshot

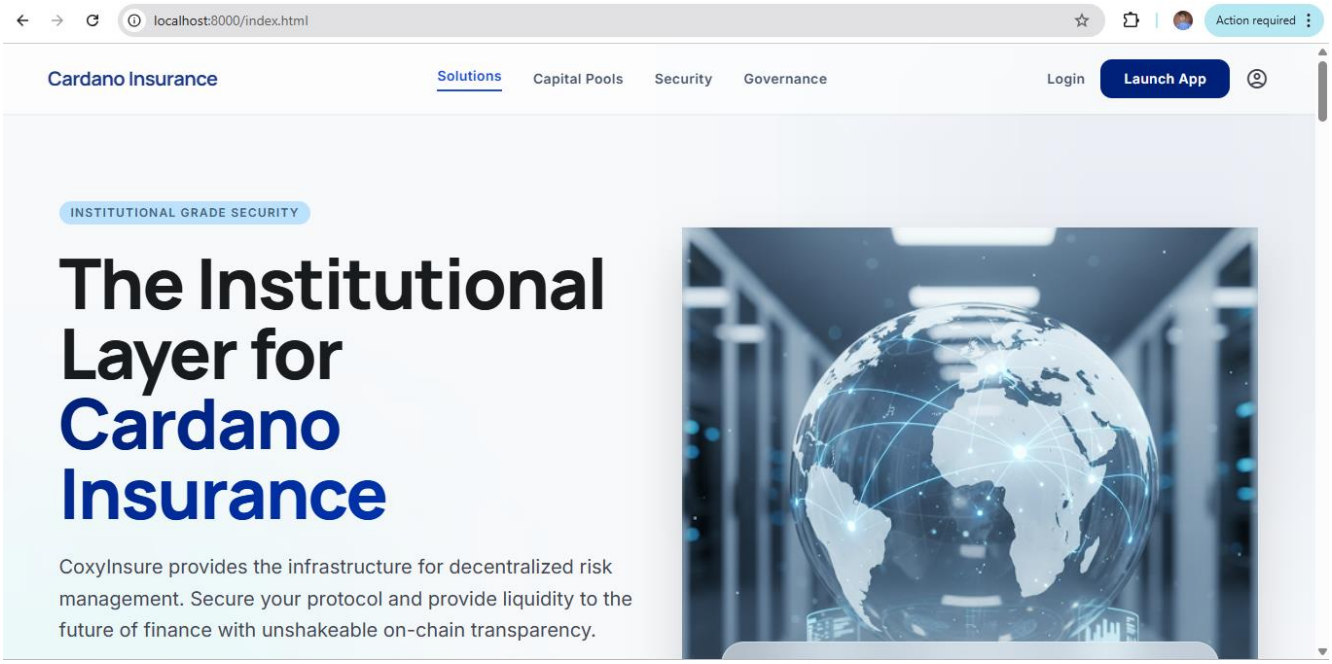

Figure A. Public entry placeholder for the CoxyInsure landing experience.

4. Global Layout and Navigation

Across major screens, the layout should stay consistent so users do not lose context while moving between tasks.

- Top header with product identity and wallet connection control
- Sidebar navigation on user dashboards for Membership, Pools, Claims, Voting, Audit, and Settings
- Primary content area divided into cards, data tables, and action sections
- Action-first layouts where forms and live tasks appear before secondary analytics
- Persistent status/log area where transaction progress and warnings are visible
- A distinct admin navigation model that mirrors the institutional dashboard style

5. Public Entry Experience

5.1 Landing / Home Screen

The landing experience should introduce the product clearly: what decentralized insurance is, who it protects, and what actions are available after connection. It should emphasize trust, institutional-grade presentation, and fast onboarding into the app.

- Headline describing Cardano-native institutional insurance value
- Short explanation of membership, risk pooling, and governance coverage
- Primary CTA to launch the application
- Feature summary or trust cards such as protocol security, auditability, and governance
- Optional metrics or social proof blocks

Placeholder B. Landing / home screen

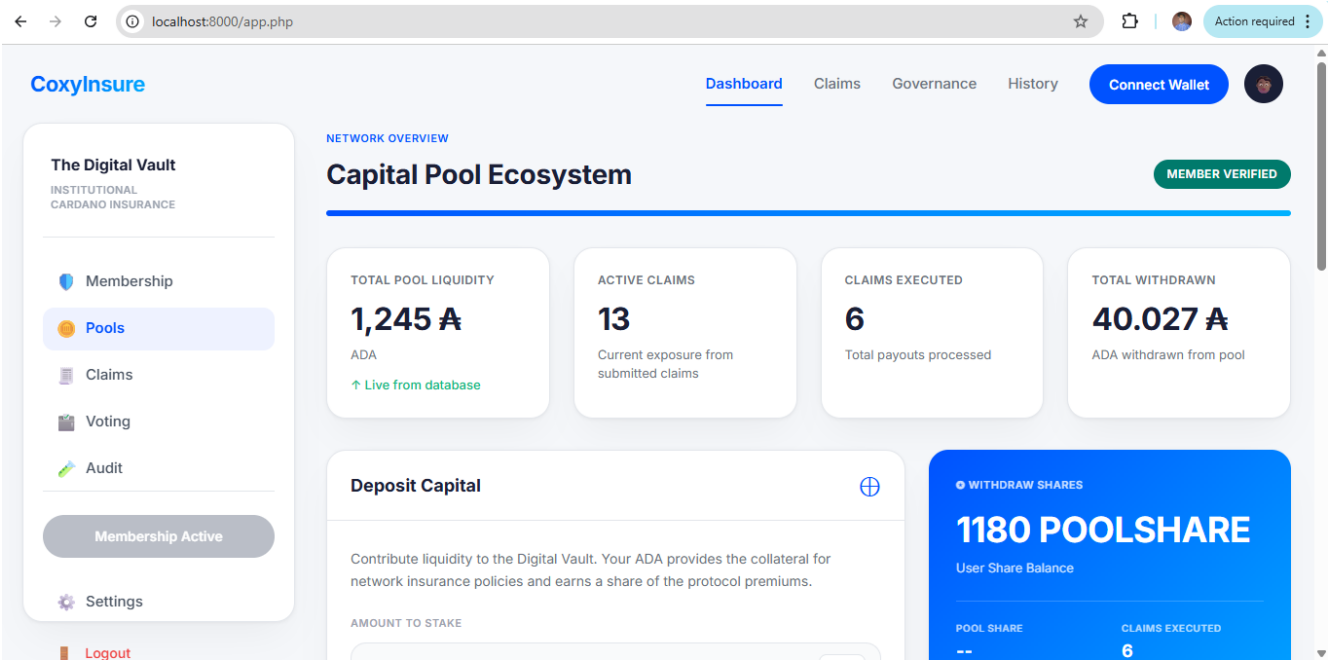

Figure B. Landing screen placeholder showing hero, trust messaging, and launch CTA.

5.2 Wallet Connection and Account Access

Wallet connection is the entry gate into the operational dApp. The design should explain when a user is connected, what wallet was connected, and whether the wallet is approved for the current account session.

- Connect Wallet button in the top header
- Wallet prompt feedback and loading state
- Account session and wallet-binding status messaging
- Wrong-wallet warning or blocked-access modal
- Successful wallet bind or admin wallet connection confirmation

Placeholder C. Wallet connection and account access

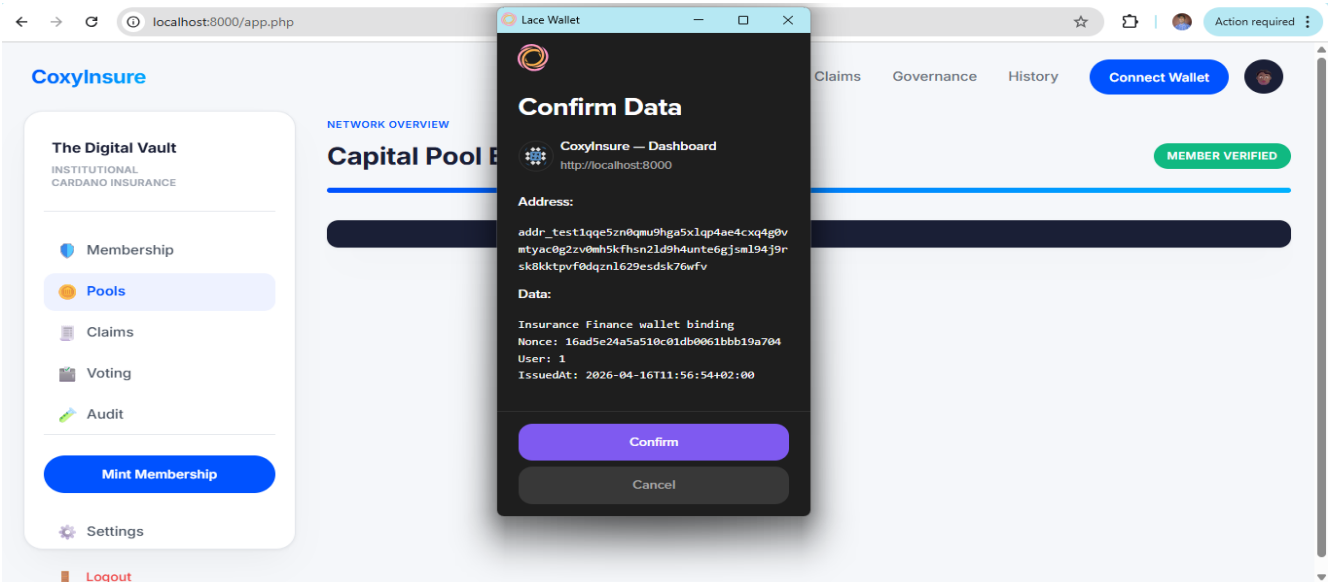

Figure C. Wallet connection placeholder for connection prompt, wallet binding, and status feedback.

6. Dashboard Overview

The user dashboard is the operational center of the product. It should present active tasks first and keep informational widgets secondary.

- Membership state or verification badge
- Capital pool ecosystem summary cards and live stats
- Deposit and withdraw controls with visible restrictions when covers are active
- Recent activity table or live protocol indicators
- Direct navigation to Claims, Voting, and Audit

Placeholder D. User dashboard overview

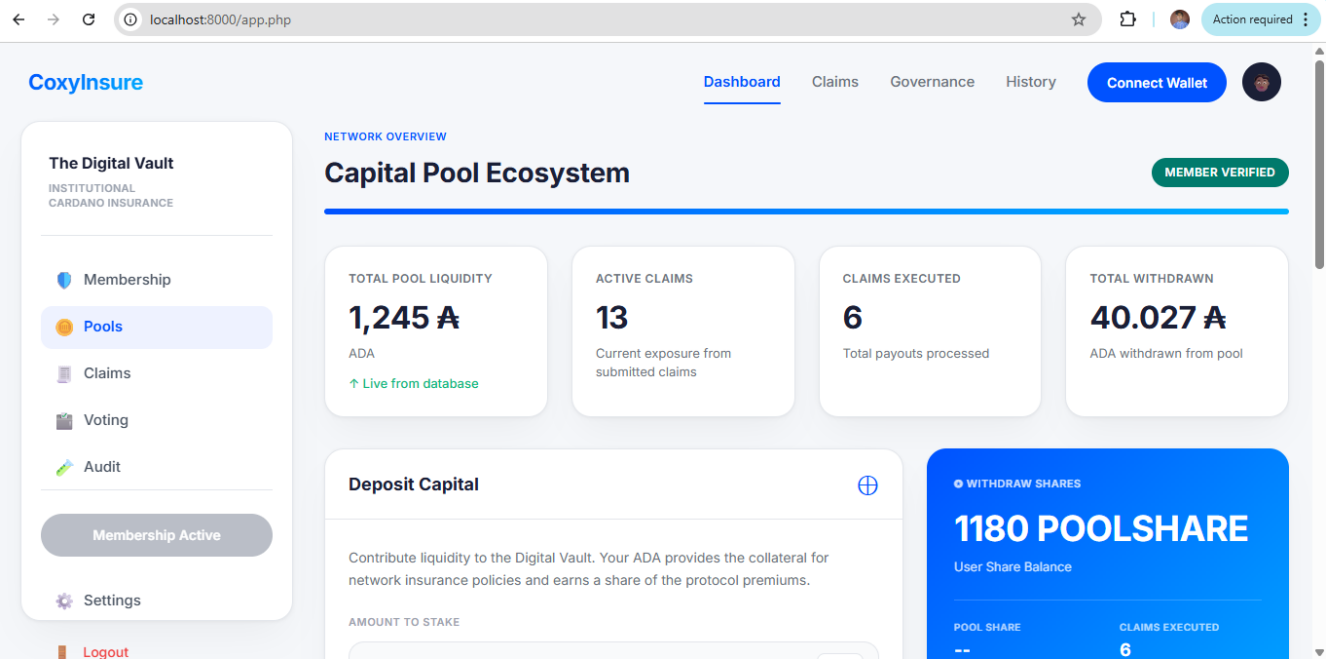

Figure D. Dashboard overview placeholder showing top navigation, stats cards, and action panels.

7. Membership Experience

Membership is the trust-entry layer for the protocol. The screen should make it clear whether the user already holds the soulbound membership NFT and whether they are eligible to continue into pool, claim, and governance actions.

- Membership-required banner when the user lacks an SBT
- Primary mint action with clear prerequisite messaging
- Post-mint verified state
- Guardrails explaining that membership is required before protected features unlock
- Status feedback when minting succeeds, fails, or is blocked

Placeholder E. Membership onboarding and verification

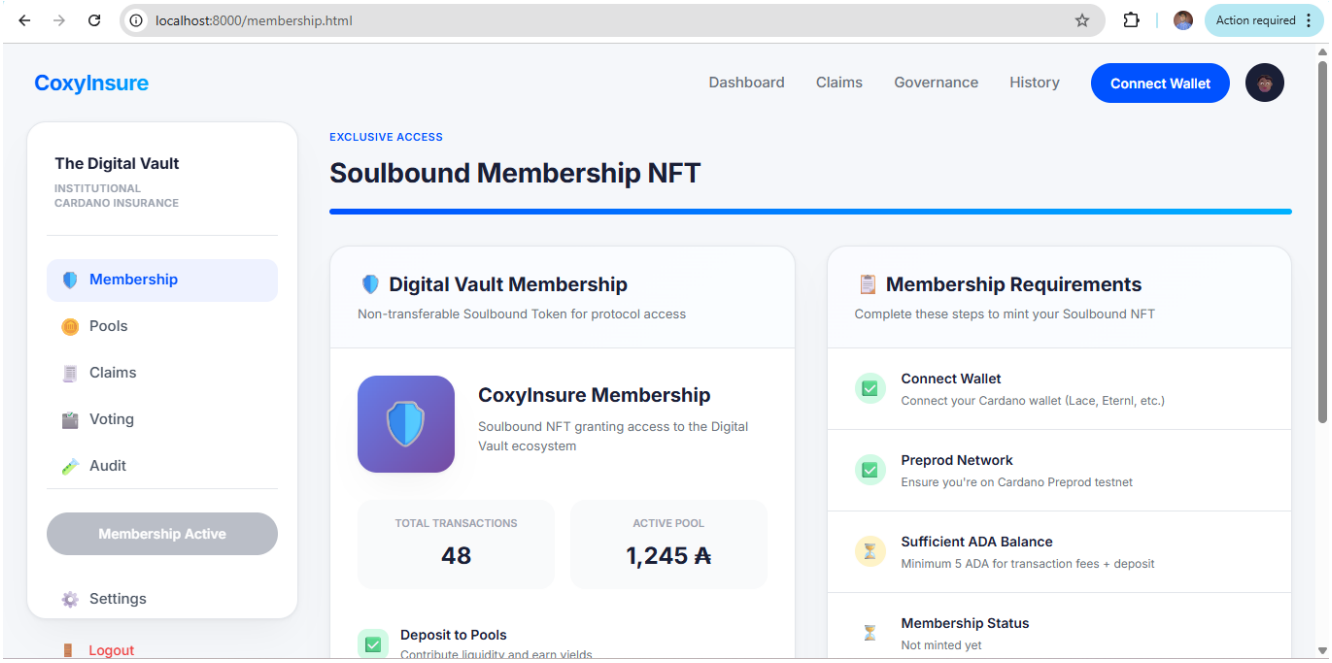

Figure E. Membership placeholder for mint state, verification state, and blocked access state.

8. Pools and Liquidity Management View

The pools page is the liquidity-provider workspace. It should support deposits, share visibility, and withdrawals while making protocol restrictions visible. Because active cover ownership can lock withdrawal privileges, the design must communicate that state clearly.

- Deposit amount input and primary deposit action
- Buy Cover shortcut adjacent to liquidity controls
- Withdraw shares card with disabled/locked state if active cover exists
- Pool stats cards and recent pool activity table
- Wallet-bound balance, share value, and recent event visibility

Placeholder F. Pools and liquidity screen

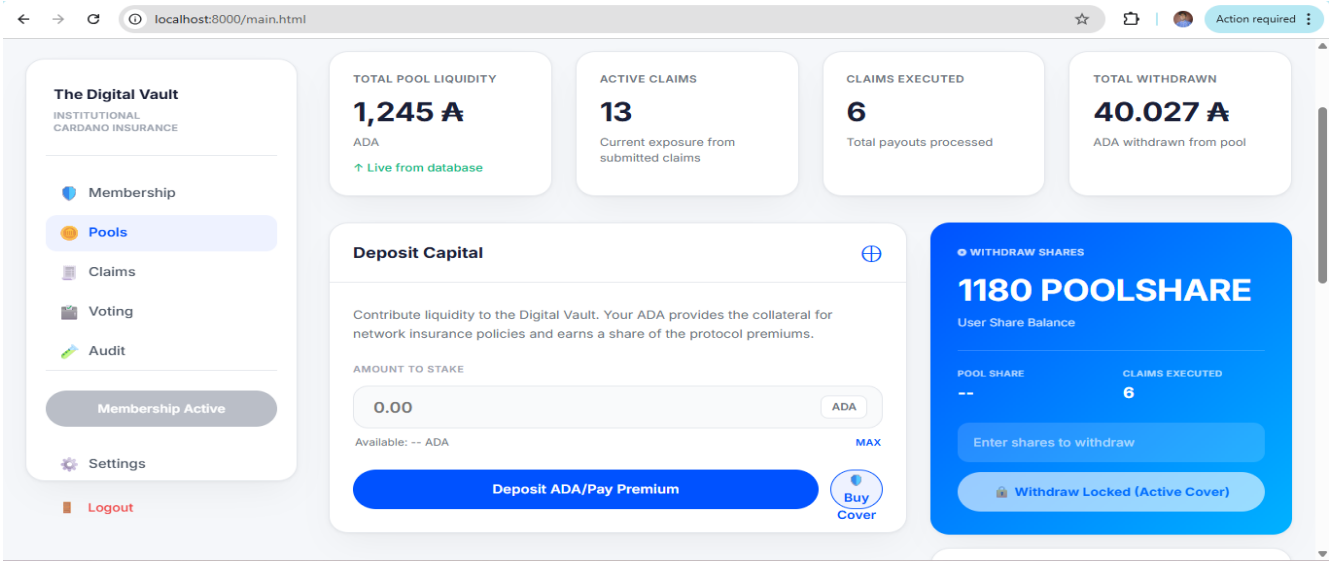

Figure F. Pools view placeholder for deposits, share balances, withdrawal lock state, and activity history.

9. Buy Cover Experience

The buy cover flow is based on fixed coverage tiers and protocol-specific durations. The interface should let users choose a protocol, see a locked duration, select a coverage tier, and purchase a cover using the displayed premium.

- Protocol selector for Minswap, Indigo, and Liqwid
- Locked duration display driven by selected protocol
- Coverage tier dropdown rather than free numeric entry
- Estimated premium block using fixed premium tiers
- Active covers panel and recent claims feed in the same screen

Placeholder G. Buy cover experience

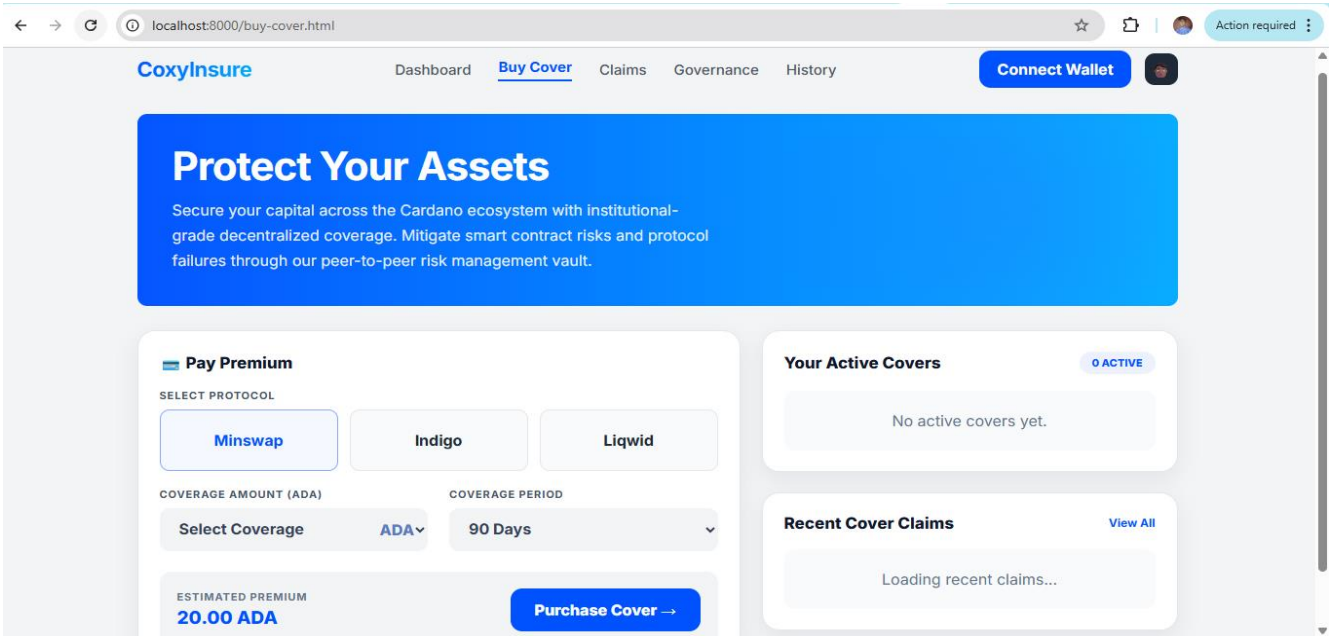

Figure G. Buy cover placeholder showing protocol tabs, tier selection, and live premium display.

10. Claims Submission and Claim History

The claims page should support a clear two-part workflow: submit a new claim with supporting evidence, and inspect wallet-scoped claim history. Evidence handling is a core trust element, so the document upload and view flow must be obvious.

- Claim amount or policy context display where relevant
- Evidence upload control with supported format guidance
- Description field and submit action
- Post-submit confirmation and transaction/log feedback
- Claim history cards listing file, size, created time, and document view action

Placeholder H. Claim submission and history

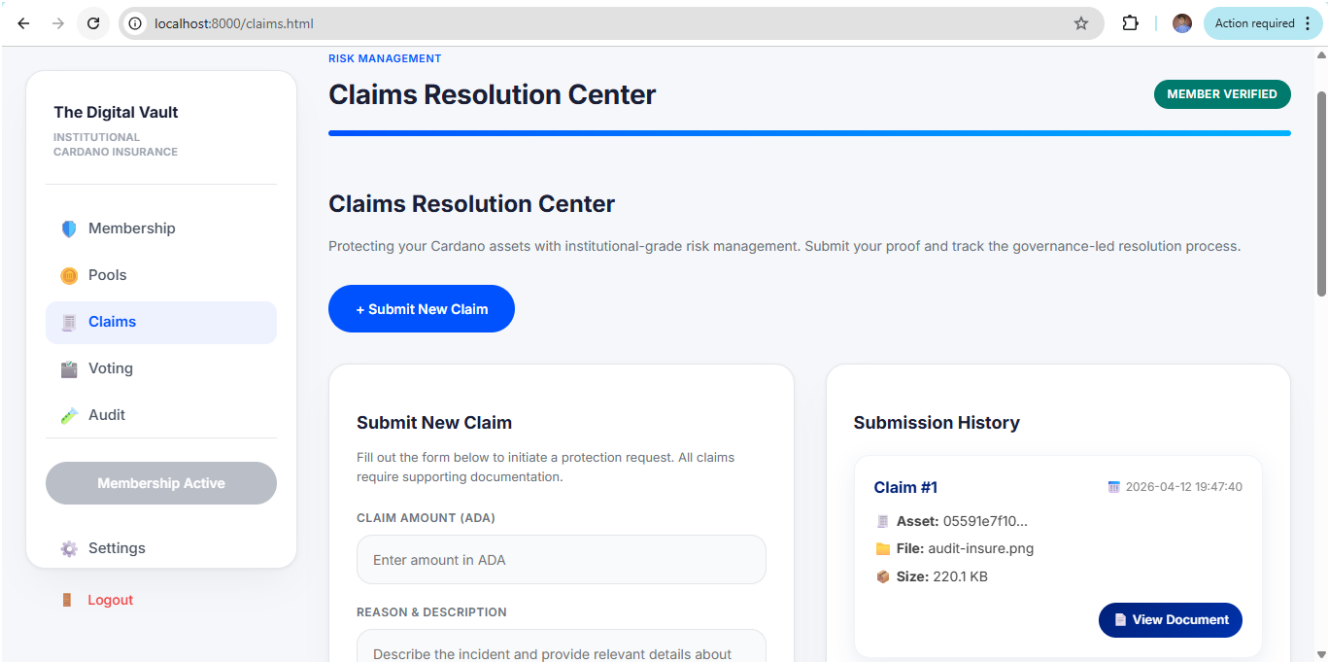

Figure H. Claims placeholder for evidence upload form and claim history cards.

11. Governance Voting and Execution

Governance is the community consensus layer. The front end should make pending votes and executable claims understandable through clear progress indicators, claimant identity display, evidence access, and threshold-based action controls.

- Pending claims grid with vote thresholds and progress bars
- Vote button with disabled state after voting
- Claim details/document access button
- Execution table listing claims that reached threshold
- Short-form claimant wallet display with copy-to-clipboard support

Placeholder I. Governance voting and execution

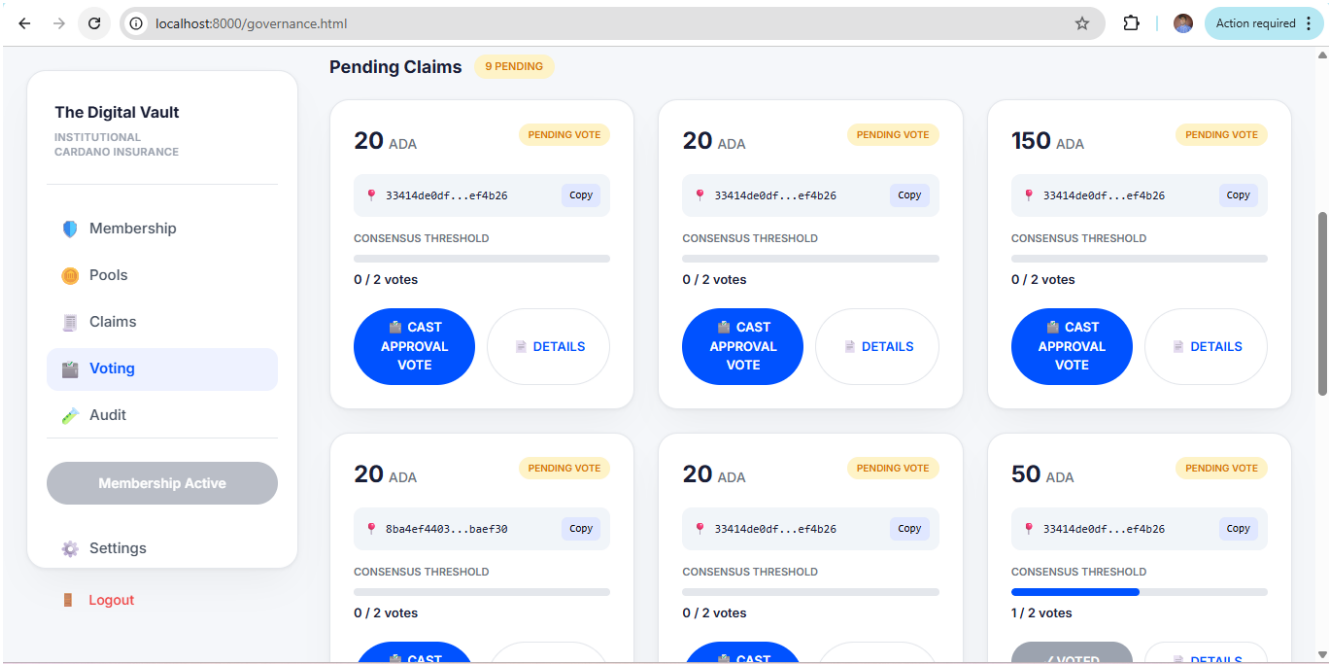

Figure I. Governance placeholder for pending votes, progress bars, execution rows, and claimant wallet copy controls.

12. Admin Front-End Surfaces

The admin portal is a separate user-facing surface with institutional styling and protected navigation. It serves monitoring, claims review, cover oversight, membership review, and governance execution duties.

- Admin login separate from end-user authentication
- Dashboard with live stats, active covers table, and recent activity
- Claims review cards with status badges
- Governance execution and rejection controls
- Membership registry and active covers search by wallet address

Placeholder J. Admin portal

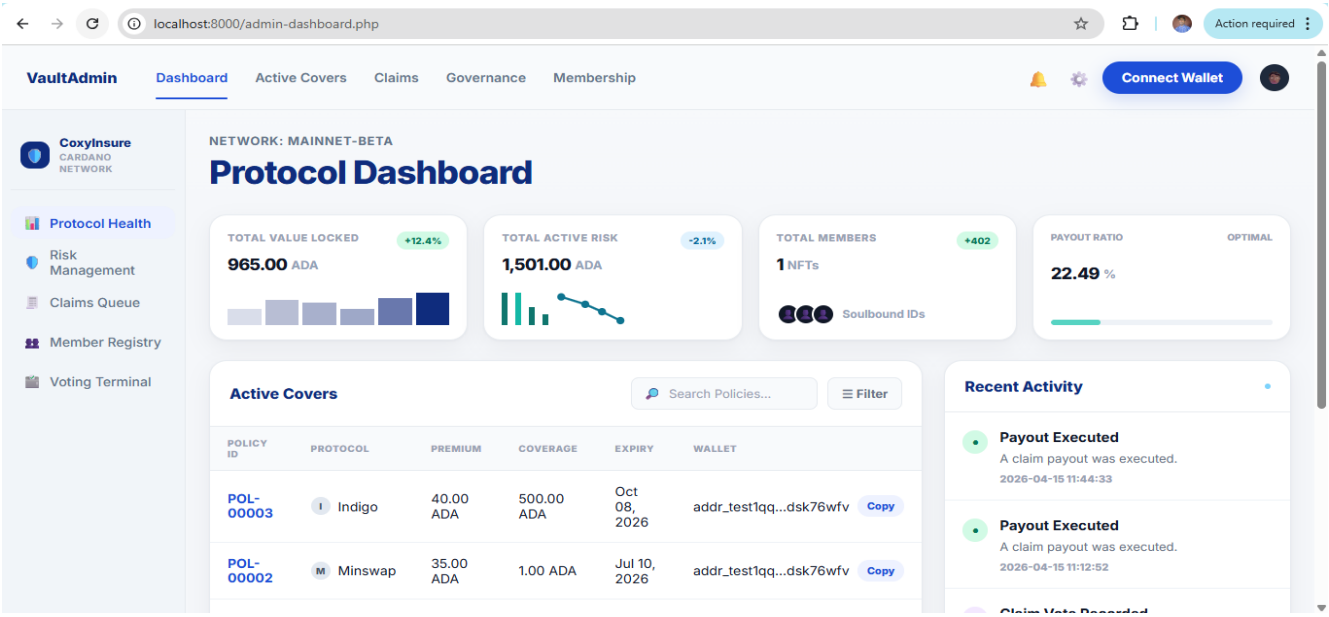

Figure J. Admin portal placeholder for dashboard, claims review, and governance control surfaces.

13. History, Audit, and Search Views

Audit and history screens provide operational transparency for both users and administrators. They should support recent activity review, transaction history tables, wallet-based searching, and export-oriented thinking even when exports are not yet exposed on every page.

- Recent transaction/activity table with strong column labels
- Wallet-address search field and clear/reset action
- Results table or cards showing matching records
- Copyable transaction identifiers, wallet addresses, or policy references
- Empty states when no records exist

Placeholder K. History, audit, and search

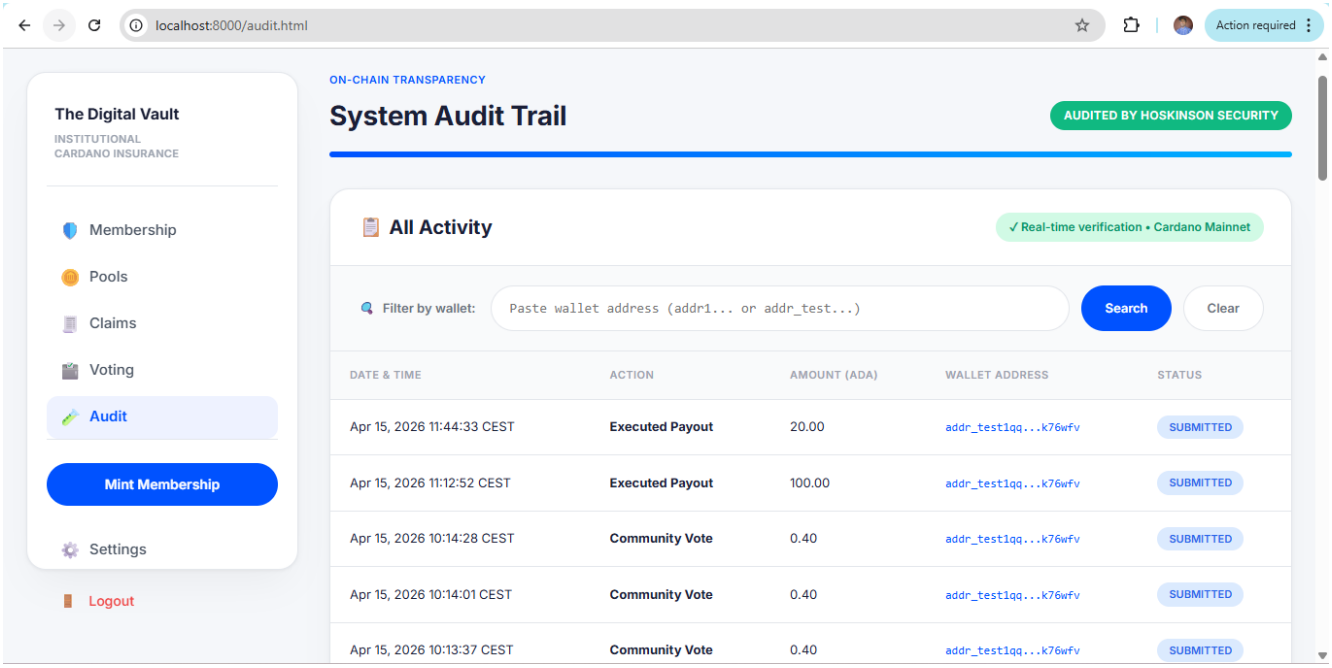

Figure K. History and audit placeholder for tables, search input, and export-oriented layout.

14. Security, Wallet Binding, and Status Feedback

Security-related design should be visible rather than hidden. Users and admins need continuous reassurance that the app is using the correct wallet, that actions are safe to approve, and that the system has enforced the right permissions.

- Verified wallet binding confirmation on user surfaces
- Wrong-wallet or blocked-access modal
- Status log area for loading, success, warning, and failure messages
- Admin wallet mode that skips end-user wallet binding but still confirms wallet connection
- Review-state awareness in governance so rejected claims disappear from public execution views

15. Reusable UI Components

- Primary and secondary buttons
- Card containers and stat cards
- Status badges for member, pending, executed, rejected, and active cover states
- Coverage tier dropdown and locked-duration field pattern
- File upload input and claim evidence pattern
- Wallet short-form chip with copy action
- Data tables, execution lists, and empty-state blocks
- Modal/dialog pattern for errors or access restrictions

16. Responsive Behaviour

- Action cards should stack vertically as horizontal space reduces.

- Grid-based areas such as cover cards and claim cards should collapse from multiple columns to one or two columns depending on width.
- Tables should preserve readability through horizontal scroll wrappers if needed.
- Buttons and copy controls should remain easy to tap and not collapse into unreadable widths.
- Sidebar layouts may shift into stacked layouts on narrower screens while keeping section hierarchy intact.

17. Accessibility and Usability Notes

- Use clear labels for all inputs rather than relying only on placeholders.
- Maintain strong contrast for brand blue, green, red, and white states.
- Provide descriptive button text for actions such as Buy Cover, Submit Claim, View Document, Execute Payout, and Reject.
- Ensure status messages remain visible long enough for users to understand what happened.
- Keep keyboard focus visible across buttons, links, inputs, and modal actions.
- Use wallet short forms carefully while still allowing full-value copy actions.

18. Future Front-End Enhancements

- Dedicated cover detail page with richer protocol metadata
- Filtered claim-review queues with claim risk flags
- Live wallet and premium compliance indicators inside governance review
- Notification center for claim updates and governance results
- Sorting and filtering for active covers, claims, and history tables
- Mobile-specific navigation pattern for core user tasks
- Claim document preview modal instead of opening a new browser tab

19. Summary

This front-end design document captures the main interaction surfaces required for the CoxyInsure DeFi insurance dApp: public orientation, wallet connection, membership verification, pools, cover purchase, claim submission, governance voting, audit review, and protected admin oversight. The layout prioritizes clarity, trust, and operational efficiency while remaining consistent with the dashboard-first pattern already established in the product.